

EXPRIMENT-4

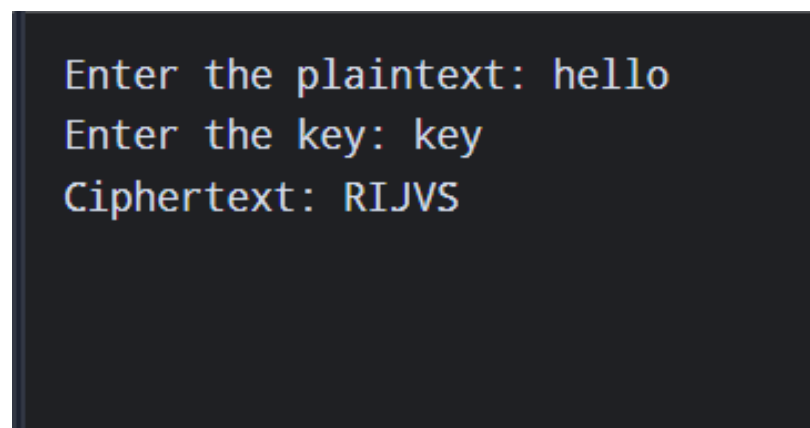
4.)Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

PROGRAM:

A screenshot of a Python IDE window titled 'main.py'. The code implements a polyalphabetic substitution cipher. It takes a plaintext and a key as input, converts them to uppercase, and then iterates through each character of the plaintext. For each character, it calculates its position in the alphabet (0-25) and adds the corresponding character from the key (also converted to its 0-25 position). The result is then converted back to a character and appended to the ciphertext. The final ciphertext is printed.

```
main.py Visualize Run ▶  
1 plaintext = input("Enter the plaintext: ").upper()  
2 key = input("Enter the key: ").upper()  
3  
4 ciphertext = ""  
5  
6 for i in range(len(plaintext)):  
7     p = ord(plaintext[i]) - ord('A')  
8     k = ord(key[i % len(key)]) - ord('A')  
9  
10    c = (p + k) % 26  
11    ciphertext += chr(c + ord('A'))  
12  
13 print("Ciphertext:", ciphertext)
```

OUTPUT:

A screenshot of the program's output. It shows the user entering 'hello' as the plaintext and 'key' as the key. The resulting ciphertext is 'RIJVS'.

```
Enter the plaintext: hello  
Enter the key: key  
Ciphertext: RIJVS
```